Matching Street View Images for the Classification of Urban Land Use From Very-High-Resolution Remote Sensing Images

Zheyao Yu, Mengmeng Li, *Member, IEEE*, and Xiangtao Zheng, *Senior Member, IEEE*

***Abstract*—Street view (SV) images provide valuable supplementary data for characterizing the functional attributes of land use types, improving urban land use classification based on very-high-resolution (VHR) remote sensing images. However, integrating SV and VHR images is challenging due to differing view modes: SV provides oblique views, while VHR offers overhead views. This discrepancy causes misalignment in both spatial and feature spaces, especially when directly mapping SV images to corresponding VHR images. To address this, we propose a novel matching method that combines spatial position constraints and viewpoint transformation. The spatial constraint establishes coarse-level correspondence, while the viewpoint transformation generates pseudo-oblique VHR images to bridge the view gap. To refine the alignment, we introduce a strip-based correlation calculation component (SCCC) that adaptively matches image features using convolutional neural networks. We then extract high-level semantic features from the matched pairs and develop an alignment-aware fusion module (AAFM) using the Sinkhorn algorithm to align their representations in feature space, enhancing semantic consistency across modalities. This, in turn, improves the classification of urban land use. We evaluated our method on two datasets, LU-FZ-VHR-SV and LU-ZZ-VHR-SV, created using GF-2 and WorldView-3 images. Results demonstrate that land use classification using the matched VHR-SV image pairs evidently outperforms classification only using remote sensing data. Comparative experiments show that our proposed SCCC and AAFM deliver superior performance over alternative correlation and fusion strategies. We conclude that integrating SV images using our approach effectively improves urban land use classification, particularly in scenes with high interclass similarity and intraclass variability.**

***Index Terms*—Feature alignment, image matching, street view (SV) images, urban land use classification, very-high-resolution (VHR) satellite images.**

I. Introduction

THE rapid pace of urbanization has led to significant population growth in cities, with more than half of the global population now residing in urban areas. This trend highlights the urgent need for sustainable urban development. In this context, detailed and updated information on urban land use is essential for policymakers to implement effective planning strategies and supporting researchers in analyzing human-environment interactions [1]. Remote sensing techniques have long been employed as effective tools for capturing the biophysical properties of the Earth's surface, supporting the extraction of land features and the classification of land types across various applications. Over the past two decades, advances in high-resolution Earth observation programs, such as the WorldView and GeoEye (USA), Pleiades (France), and GaoFen (China) satellites, have significantly increased the availability of high-resolution images. This abundance of data offers great potential for detailed land information extraction. In particular, very-high-resolution (VHR) images, i.e., spatial resolution finer than 1 m, have demonstrated considerable promise in identifying the functional types of land, supporting urban land use classification via extracting high-level image features that abstract human visual interpretation [2], [3].

Many studies have investigated the use of VHR images for urban land use classification [4], [5]. A straightforward approach involves using machine learning classifiers with manually engineered features to identify urban land use types [6]. With the rapid advancement of deep learning in remote sensing, deep neural networks have demonstrated remarkable capability to automatically extract high-level image features [7], [8], [9]. For example, Huang et al. [10] proposed a convolutional neural network-based framework that extracts fine-grained local spatial features from images using a pre-trained model, and further fine-tuned the extracted features through additional convolutional layers to generate robust feature representations for improving the accuracy of urban land use classification. Inspired by the transformer's ability to model long-range spatial dependencies, Bazi et al. [11] introduced a vision transformer (ViT)-based approach for VHR image classification [12]. Their results showed that the ViT-based model outperformed traditional CNN-based methods in classification accuracy. However, they also noted that self-attention-based models lack inherent spatial inductive biases, making it difficult to capture fine-grained local features without additional mechanisms. To address this limitation, recent studies have explored the integration of CNNs and transformers, enabling the simultaneous extraction of local spatial details and modeling of global contextual relationships [13]. Despite these advances, urban land use units often contain diverse land cover types, leading to high interclass

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Zheyao Yu and Mengmeng Li are with the Key Laboratory of Spatial Data Mining and Information Sharing of Ministry of Education, Academy of Digital China (Fujian), Fuzhou University, Fuzhou 350108, China (e-mail: 235527062@fzu.edu.cn; mli@fzu.edu.cn).

Xiangtao Zheng is with the College of Physics and Information Engineering, Fuzhou University, Fuzhou 350108, China (e-mail: xiangtaoz@gmail.com).

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similarity and intraclass variability in VHR images. As a result, distinguishing certain urban land use categories, such as commercial and industrial use, remains challenging when relying solely on VHR images [14].

To enhance the separability of challenged urban land use types, previous studies have recognized the importance of integrating different sources of data with VHR images for urban land use classification. For example, Liu et al. [58] employed a transformer-based architecture to separately extract global spatial features from VHR images and polarimetric features from SAR data, in which scattering differences under various polarization modes capture the geometric structure and material properties of surface objects. By fusing data at both the feature and decision levels, they further improved classification accuracy. Feng et al. [16] enhanced the recognition of urban land use types with limited training samples (e.g., playgrounds and parking lots) by combining fine-grained spatial features from VHR images (extracted via CNNs) with height-related features derived from LiDAR data. Nevertheless, the multisource VHR, SAR, and LiDAR data all capture urban scenes from an overhead view. While such data provide detailed information about rooftops and building heights, they offer limited insight into building facades and street scenes, making it difficult to differentiate land use types in complex urban environments [17], [18], [20]. In this context, street view (SV) images captured from oblique view offer a more natural visual representation of the urban environment, providing supplementary functional information for the characterization of urban land use types [20], [21]. A typical approach for integrating SV and VHR images involves two main steps. First, features are independently extracted from the VHR image and its corresponding SV image, e.g., a nearby SV image [22]. Second, the extracted features from both sources are then fused for image classification. However, this approach fails to considering the spatial projection of the street elements in an SV image onto the corresponding remote sensing image. In practice, the field of view (FOV) of SV images can vary significantly. When the FOV is excessively wide, a single SV image may span multiple land use units, resulting in mixed land use information within the image. This mixture can reduce classification accuracy by introducing ambiguity in the correspondence between SV content and the actual land use type [23].

Furthermore, differences in view modes between VHR and SV images complicate the correspondence at the pixel level, posing significant challenges for spatial alignment [24], [25], [26], [27]. To deal with this challenge, Workman et al. [28] proposed a geospatial attention mechanism that calculates the Haversine distance and orientation angle between the acquisition point of each SV image and every pixel in the VHR image. This approach enables the identification of relevant regions in the remote sensing image that are spatially associated with the corresponding SV image. Their method, however, fails to consider the geometric projection differences between oblique and overhead view modes. Recently, Li et al. [29] proposed a geometry-based ray tracing matching method, and incorporated north direction reference information from SV images to determine the correspondence between buildings in the SV images and buildings in the VHR images. This method, however, relies on access to accurate north direction data for SV images, which limits its applicability. Since many SV image providers do not supply north direction information, establishing reliable matching between SV and VHR images remains a significant challenge.

This study aims to develop a novel method for automatically matching SV images with VHR images to enhance urban land use classification, addressing misclassification issues arising from the insufficiency of VHR images in characterizing functional properties. The proposed matching method first simulates the initial orientation of the SV images during capture based on road networks. This step models the coarse-level spatial correspondence between the limited FOV SV images and the VHR images. Next, it accounts for the geometric projection differences between the pair of VHR-SV images and adjusts the view mode of the VHR images accordingly, allowing deep neural networks to learn the paired image features and achieve fine-level spatial correspondence. It is noteworthy that the human visual system typically employs a hierarchical processing approach to recognize images [30]. Based on this idea, we introduce a correlation calculation component based on a strip-layered strategy to reduce the interference from the complexity of urban environments in image matching. Previous studies have shown that reinforcing interfeature correlations, using the Sinkhorn algorithm, prior to feature fusion leads to more robust representations [31]. This motivates us to integrate the Sinkhorn mechanism into a cross-attention module to enhance the fusion of matched image pairs and mitigate misclassifications caused by view-mode differences. The main contributions of this study are as follows.

1) It proposes a novel image matching method to align SV images with corresponding VHR aerial images using spatial constraints and open road networks. The method first estimates the initial orientation of the SV image to narrow the search space during matching. A viewpoint transformation is then applied to address the inherent view-mode discrepancy between SV and VHR images.
2) It introduces a new strip-based correlation calculation component (SCCC) to dynamically compute multilevel semantic correlations between VHR-SV image pairs. By using a strip-layered strategy inspired by attention mechanisms, SCCC effectively captures both local and global correlation scores, enabling fine-grained matching and reducing errors in complex urban scenes.
3) It develops an alignment-aware fusion module (AAFM) using the Sinkhorn algorithm to align and fuse semantic features from the VHR-SV image pairs. This module optimizes the distribution of objects from the same category across different view modes in the feature space, enhancing the characterization of land use functions.

The remainder of this article is organized as follows. Section II describes the study areas and datasets. Section III illustrates the proposed method for image matching and urban land use classification. Experimental results and related analysis are presented in Section IV, followed by a discussion in Section V and conclusions of this research in Section VI.

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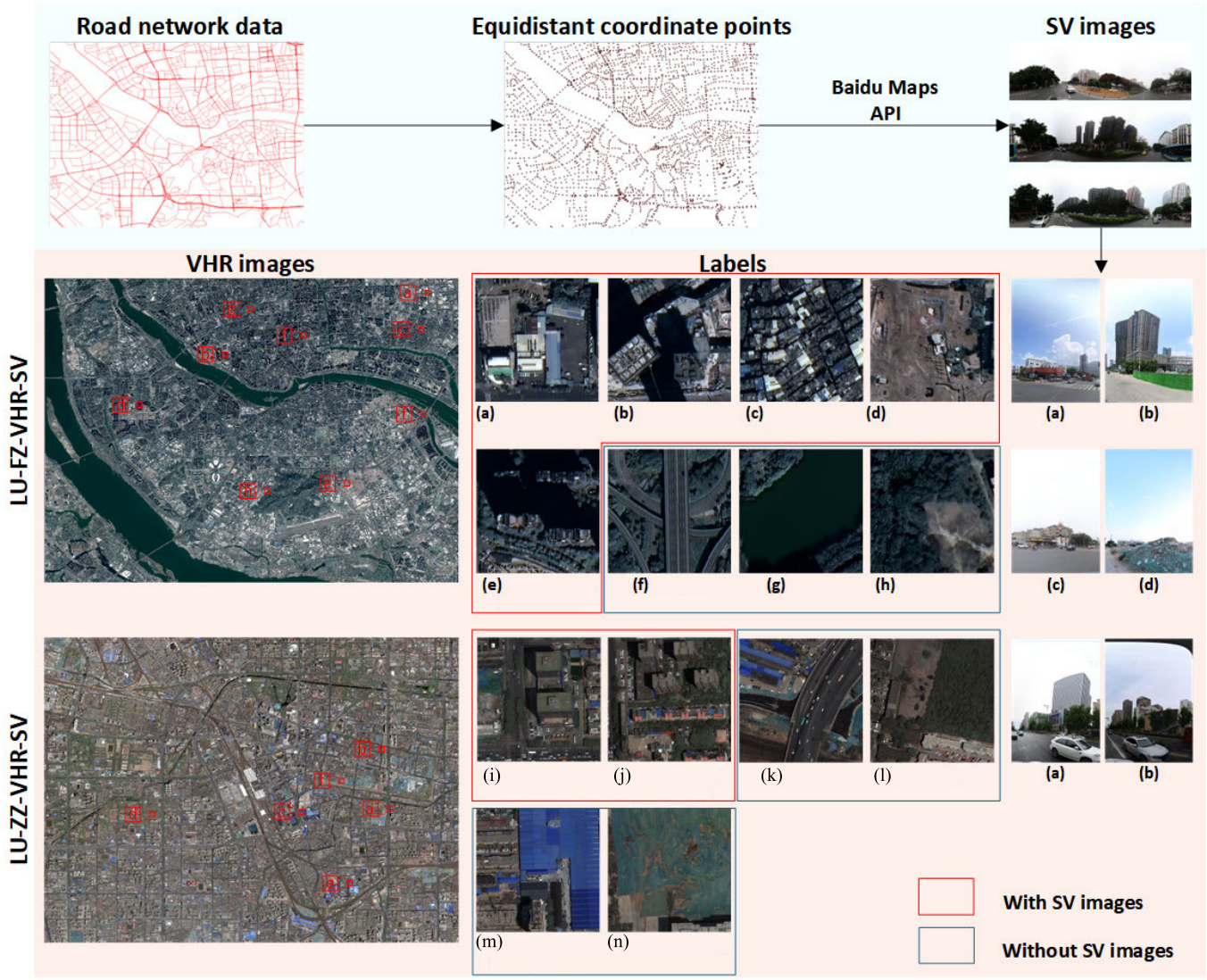


Fig. 1. Overview of the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets, combining VHR images with the corresponding SV images. (a) and (m) Industrial and warehouses, (b) and (i) commercial, (c) high-density residential, (d) and (n) undeveloped land, (e) and (j) low-density residential, (f) and (k) others, (g) water, and (h) and (l) green space.

II. Study Areas and Datasets

We tested our method on two new datasets consisting of SV images and VHR images in Fuzhou (FZ) and Zhengzhou (ZZ) cities, China (see Fig. 1). FZ and ZZ are the capitals of Fujian and Henan provinces in China, respectively, featuring diverse urban structures and street layouts among rapid urbanization. More specifically, the FZ study area captured Chinese GaoFen-2 VHR images, with one panchromatic of spatial resolution 0.78 m, and four multispectral bands of 2.5 m, while the ZZ study area used the US WorldView-3 VHR images with one 0.3-m panchromatic band and four 1.2-m multispectral bands. We conducted pansharpening to fuse the multispectral and panchromatic bands for each image and resampled the pansharpened images to 0.8 and 0.3 m for the two areas, respectively.

We obtained road network data for the two areas from https://www.openstreetmap.org. We first preprocessed the data to remove road segments with extremely low traffic flow, and to eliminate poorly connected and less significant segments based on the analysis of road connectivity. We then placed equidistant coordinate points on the road networks at 100-m intervals and acquired panoramic SV images of each point from https://lbsyun.baidu.com/index.php?title=webapi. The input VHR images were cropped into patches with 448×448 pixels centered around the equidistant coordinate points. Next, each obtained SV image was further divided into four equal parts with 512×664 pixels for each part, while the cropped image patch was divided into four parts with 224×224 pixels for each part. In this study, we constructed two datasets for the two study areas, labeled LU-FZ-VHR-SV and LU-ZZ-VHR-SV, which combine the paired SV and VHR images for each area.

In the LU-FZ-VHR-SV dataset, VHR-SV image pairs were annotated into five land use classes: industrial and warehouse, high-density residential, low-density residential, commercial areas, and undeveloped land. In addition, three more classes, i.e., water bodies, green spaces, and other land uses, were annotated from VHR images only, as these categories are

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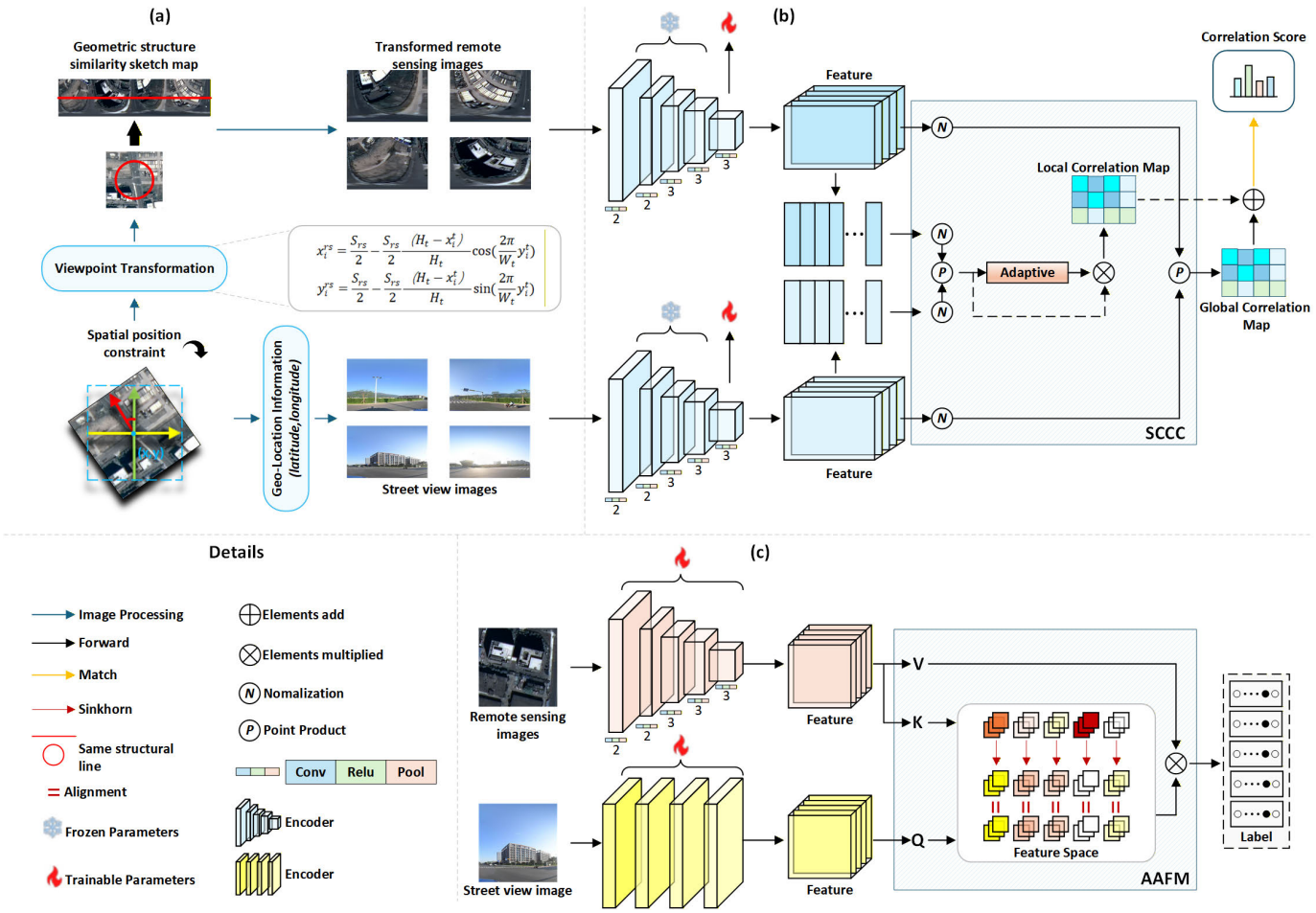


Fig. 2. Workflow of the proposed method for urban land use classification. It comprises three main steps: (a) viewpoint alignment, (b) image matching, and (c) image classification. SCCC and AAFM refer to the strip-based correlation calculation component and alignment-aware fusion module, respectively.

difficult to discern from the oblique perspective of SV images. The LU-ZZ-VHR-SV dataset includes VHR-SV pairs annotated into two classes: low-density residential and commercial lands. Annotations from VHR-only images cover four more classes: industrial and warehouse, undeveloped land, green spaces, and other land uses. The classification system was adapted from the Chinese National Land Use Classification Standard (GB/T 21010-2017) and tailored for urban land use analysis [32], [33], [34]. For each category, 200 samples were selected and split into training and validation sets in an 8:2 ratio. All annotations were derived through visual interpretation, supported by points of interest (POI) data from Baidu Maps [34]. These datasets are available at: https://www.github.com/diolock777/LU_VHR_SV_Datasets

III. Methodology

The workflow of the proposed urban land use classification is illustrated in Fig. 2. First, road networks are used to impose spatial constraints on each VHR-SV image pair at every SV capture point, establishing coarse-level spatial correspondence. To address differences in view modes, a viewpoint transformation converts the VHR image into a pseudo-oblique view, making it more comparable to the corresponding SV image. Next, an SCCC is developed to dynamically enhance fine-level matching by computing correlations between high-level features of the image pairs. To effectively fuse the matched VHR and SV features, we design an AAFM. This module strengthens the correlation between query and key features, aligning feature spaces to reduce misclassification caused by view modes' differences. Finally, the fused features are used for urban land use classification.

A. Aligning SV and VHR Images in Space

1) Establishing Coarse Spatial Correspondence for Vhr–Sv Paired Images Under Spatial Position Constraints: When the position and orientation of SV images are available, spatial alignment with VHR images can be achieved in a straightforward way. As shown in Fig. 3, each SV image is a 360° panorama, with the vehicle's driving direction serving as the reference direction. The panoramic SV image is divided into four 90° segments, representing the front-left, front-right, back-left, and back-right views relative to the vehicle's heading [see Fig. 3(b)]. Each panoramic SV image corresponds to a 448 × 448 pixel VHR image. Then, we divided the 448 × 448 VHR image evenly into four 224 × 224 pixel image quadrants along the vehicle's driving direction, corresponding to the front-left, front-right, back-left, and back-right views of the vehicle's driving direction. This division

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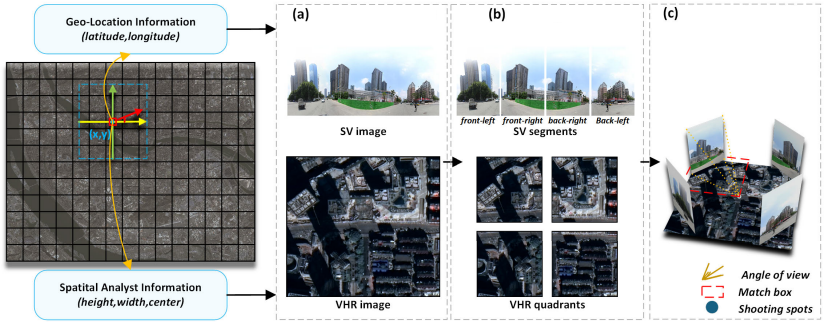


Fig. 3. Illustration of the coarse spatial correspondence between SV and VHR images. (a) Input the SV image and its corresponding VHR image. (b) SV image is divided into four directional segments, i.e., front-left, front-right, back-left, and back-right, based on the vehicle's orientation, while the VHR image is partitioned into four corresponding quadrants. (c) Visualization of the established correspondence between each SV segment and its associated VHR quadrant.

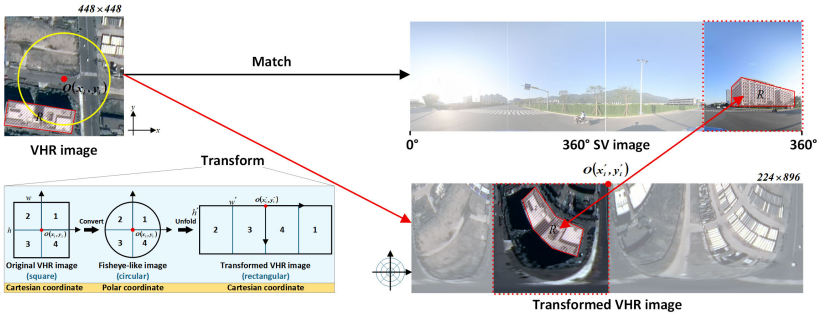


Fig. 4. Illustration of the viewpoint transformation process. A fisheye-like circular image is first generated to capture the 360° surroundings of an SV capture point within a VHR image. This circular image is then unfolded into a rectangular representation, resulting in the transformed VHR image.

allows each SV segment to be automatically matched with its respective VHR quadrant [see Fig. 3(c)].

However, the vehicle's driving direction is not always available and may not be located horizontally or vertically in the image space. To address these limitations, we determine the vehicle's driving direction based on road networks, by assuming that a vehicle captures SV points along road centerlines no matter it drives forward or backward and conduct a rotation operation for each 448 × 448 pixel VHR image. Specifically, based on this inferred direction, we rotate the corresponding VHR image by a specific angle, denoted as α. Let $o(x, y)$ represent the centroid of rotation, corresponding to the SV capture point. To calculate the rotation angle α, we construct a directional line segment centered at $o(x, y)$ and extending a fixed distance d (i.e., $d = 10$ pixels in this study) in both directions along the road centerline. This yields two endpoints: $o_1(x_1, y_1)$ and $o_2(x_2, y_2)$. The rotation angle α is then calculated as

$$\alpha = \frac{180}{\pi} \cdot \operatorname{atan2}(y_2 - y_1, x_2 - x_1). \tag{1}$$

After rotating the VHR image by angle α, we divide it into four VHR quadrants. This quadrant-based division enables a coarse spatial correspondence between the SV image segments and the respective regions in the rotated VHR image.

2) Eliminating View-Mode Differences via Viewpoint Transformation: To address the inherent view-mode discrepancy between SV and VHR images, i.e., the oblique perspective of SV images versus the overhead perspective of VHR images, we apply a viewpoint transformation that converts the VHR image into an oblique-like view resembling that of the SV image. This results in a transformed VHR image that more closely aligns with the SV modality, thereby reducing geometric inconsistencies and facilitating improved alignment in the feature space. The motivation behind this transformation is illustrated in Fig. 4. Considering a building object R as an example, the SV image captures a 360° panoramic view of its surroundings, which corresponds to a circular region centered at the SV capture point $o(x, y)$ in the VHR image. The radius of this circular region is equal to the distance between the object R and the capture point $o(x, y)$, as indicated by the yellow circle in Fig. 4. To simulate the SV viewing geometry, we first generate a fisheye-like or "circular" image that captures the 360° surroundings relative to the center. We then unfold the circular image into a rectangular representation, i.e., the transformed image, based on the transformation rules defined in [35]. This polar transformation maps VHR features into a panoramic view like SV images, thereby narrowing the view-mode gap between the two modalities and enhancing their compatibility in downstream tasks.

Let H and W be the height and width of a VHR image, and H' and W' be the transformed height and width. Given a pixel coordinate (x_i, y_i) of the VHR image, its transformed coordinate (x'_i, y'_i) is related as follows:

$$x'_i = \frac{W}{2} - \frac{H}{2} \cdot \frac{H' - x'_i}{H'} \cdot \cos\left(\frac{2\pi}{W'} \cdot x'_i\right) \tag{2}$$

$$y'_i = \frac{H}{2} - \frac{W}{2} \cdot \frac{H' - x'_i}{H'} \cdot \sin\left(\frac{2\pi}{W'} \cdot y'_i\right) \tag{3}$$

where $(H' - x'_i)/H'$ acts as the scaling factor, normalizing the pixel coordinates in the VHR image relative to the height of

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the transformed image. The $\cos(2\pi/W' \cdot x_i')$ and $\sin(2\pi/W' \cdot y_i')$ serve as rotation factors. They adjust the VHR pixel coordinates according to the angular position in the circular (polar) domain, effectively mapping them to horizontal and vertical directions in the original image.

This viewpoint transformation reduces the spatial differences between VHR and SV images, achieving coarse matching between them in this study.

B. Fine Matching via a Dynamic Image Correlation Adjustment

1) Matching Method: After minimizing the spatial differences between VHR and SV images, we compare their feature correlation for fine-grained matching. Specifically, we first extract high-level features from both SV and transformed VHR images using a modified VGG16 network [36], to ensure semantic consistency. Next, we retain the first four convolutional blocks of VGG16, freezing the parameters of the first three blocks (i.e., the initial ten layers) to preserve low-level feature consistency between the two image types. Following the fourth block, we introduce a customized convolutional module comprising three additional convolutional and pooling layers to enhance the learning of high-level representative features while maintaining semantic alignment.

These extracted features are then used to calculate the correlation between the image pairs. To better capture local spatial details, we develop an SCCC, where correlations are calculated as point products between corresponding feature pixels, resulting in correlation scores (see Fig. 5). An attention mechanism [37] is subsequently applied to dynamically refine these scores, enabling more accurate fine-grained image matching. Specifically, let $F_{\text{pt}} \in \mathbb{R}^{h\times w\times c}$ and $F_{\text{sv}} \in \mathbb{R}^{h\times w\times c}$ denote the extracted features from the transformed VHR and SV images, respectively, where h, w, and c represent the height, width, and number of channels. Each feature pixel is located as $x_{\text{pt}}(h,w,c)$ and $x_{\text{sv}}(h,w,c)$. We first normalize the feature maps F_{pt} and F_{sv}, and then calculate their point product at each pixel to obtain pixel-level correlation. We then slice the two normalized features along the width dimension into N equally spaced stripes, where the i-th stripe is denoted as F_{pt}^i and F_{sv}^i, which have a width of W/N. We can calculate the stripe correlation scores s_{stripe}^i between the two image stripes, defined as follows:

$$s_{\text{stripe}}^i = F_{\text{pt}}^i(c,h,w) \cdot F_{\text{sv}}^i(c,h,w). \tag{4}$$

To estimate the relative importance of different stripes in the correlation calculation, we define a vector $\boldsymbol{s_{stripe}} = (s_{\text{stripe}}^1, s_{\text{stripe}}^2, \ldots, s_{\text{stripe}}^N)$, which stacks the correlation scores from all N stripes. A linear layer is then applied to transform s_{stripe}, followed by a softmax operation to obtain the attention weights α, representing the contribution of each stripe. The weight calculation is given by

$$\boldsymbol{\alpha} = \text{softmax}\left(W_1 \cdot \boldsymbol{s}_{\boldsymbol{stripe}}^T + \boldsymbol{b}\right) \tag{5}$$

where W_1 denotes a weight matrix and $\boldsymbol{b}$ is a bias vector, both of which are the learnable parameters of the linear layer. The softmax ensures that the attention weights sum to one. To achieve a more comprehensive representation of feature correlation, we first calculate the local correlation scores by performing a weighted sum of the strip correlation scores using the elements α_i in the weight $\boldsymbol{\alpha}$. Then, we sum all the strip correlation scores to derive the overall correlation score. Finally, we introduce an adaptive adjustment parameter λ to adaptively balance the contributions from local and global correlation scores. The final correlation score s_{final} is obtained as follows:

$$s_{\text{final}} = \lambda \cdot \sum_{i=1}^{N} s_{\text{stripe}}^i + (1-\lambda) \cdot \sum_{i=1}^{N} \alpha_i \cdot s_{\text{stripe}}^i. \tag{6}$$

By analyzing the value of s_{final}, we can determine the matching relationship between the transformed VHR and SV images.

2) Loss Function for Matching: Our method involves the correlation calculation of matching relationships, so a triplet loss function is employed to optimize the correlation relationships between the transformed VHR and SV images [38]

$$\mathcal{L} = \max\left(0, \text{Mar} + s_{\text{final},n} - s_{\text{final},p}\right) \tag{7}$$

where $s_{\text{final},n}$ represents the final correlation scores between nonmatching samples, $s_{\text{final},p}$ represents the final correlation scores between matching samples, and Mar is a hyperparameter that controls the margin between the correlation of positive and negative samples. Moreover, to enhance the diversity of triplets, a detailed mini-batch strategy was adopted to construct training triplets [39], [40], [41]. Each mini-batch contains four pairs of transformed VHR images and SV images. For each

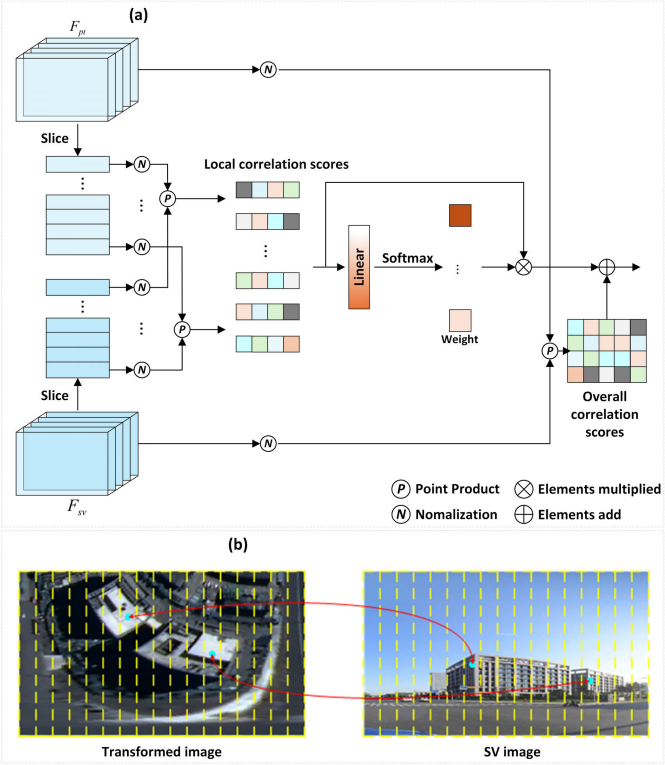


Fig. 5. (a) Architecture of the proposed SCCC. It enhances correlation computation between the transformed VHR feature and the SV feature. (b) Illustration of the strip partitioning strategy, where red lines indicate potential positional correspondences between the two images.

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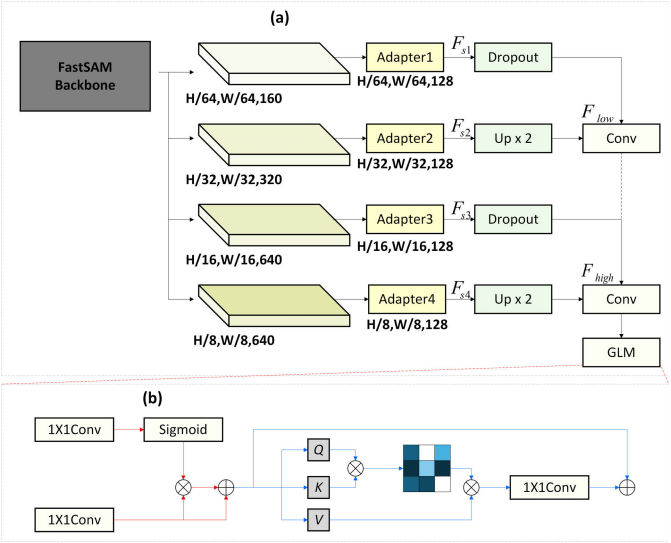


Fig. 6. Architecture of (a) SV images feature extraction branch and (b) GLM.

transformed VHR image, one matching SV image (positive sample) and three nonmatching SV images (negative samples) are selected. All four images are acquired at the same equidistant coordinate point. This construction improves the robustness of the model's learning.

C. Urban Land Use Classification Based on Matched Image Pairs

In this study, we construct a novel classification method using the matched VHR-SV image pairs for urban land use classification. Our classification method consists of three key components: SV image feature extraction, VHR image feature extraction, and feature fusion. In SV image feature extraction, to mitigate background interference and capture task-relevant information, we employ FastSAM to extract multilevel semantic features from the SV images. Furthermore, a global–local feature enhancement module (GLM) is introduced to effectively integrate the multilevel semantic features. In VHR image feature extraction, VGG16 is selected as the backbone due to its hierarchical convolutional structure, which is capable of effectively extracting the high-level semantic features [43]. To address the feature misalignment caused by the distinct view modes of VHR and SV images and to achieve effective fusion, we propose an AAFM. This module aligns semantic features in the feature space and fuses them, thereby improving the accuracy of urban land use classification.

1) Feature Extraction of Vhr-Sv Images: SV images often contain complex urban features and are affected by factors such as occlusion and lighting variations, making them challenging for traditional feature extraction methods. FastSAM, a variant of segment anything model (SAM), retains the feature extraction capabilities of SAM trained on large datasets, while significantly reducing computational overhead [42], [44]. In this study, we use FastSAM to extract multilevel semantic features from SV images. The multilevel semantic features extracted from FastSAM's output layer are sequentially defined from low to high as follows: F_1, F_2, F_3, and F_4 (see Fig. 6). To normalize multiscale semantic features, we uniformly mapped these features to the same number of channels using a convolutional adapter. Each convolutional adapter consists of a 1×1 convolution layer and a batch normalization. The normalized feature F_{si} is obtained by

$$F_{si} = \text{Adapter}(F_i), \quad i = 1, \ldots, 4. \tag{8}$$

These normalized features are further mapped to the same spatial resolution through upsampling. As illustrated by Fig. 6(a), we first upsample the F_{s2} and sum it with F_{s1} to generate low-level semantic features F_{low}. Then, we upsample F_{s4} and sum it with F_{s3} to produce high-level semantic features F_{high}. The computation is as follows:

$$F_{\text{high}} = F_{s3} + \text{Upsample}(F_{s4}) \tag{9}$$

$$F_{\text{low}} = F_{s1} + \text{Upsample}(F_{s2}). \tag{10}$$

Moreover, inspired by Jiang et al. [45], we developed a GLM to integrate low-level and high-level semantic features to get the enhanced features [see Fig. 6(b)], defined as

$$F_{\text{SV}} = \sigma\left(F_{\text{high}}\right) + \sigma\left(F_{\text{high}}\right) \cdot \delta\left(\sigma\left(F_{\text{low}}\right)\right) \tag{11}$$

where F_{SV} refers to the integrated features of SV images, $\sigma(\cdot)$ represents a 1×1 convolution, and $\delta(\cdot)$ represents the sigmoid function. Next, we use a self-attention component to model global contextual dependency among SV images, improving the identification capability of our model for complex scenarios [46]. Specifically, three parallel 1×1 convolutions are used to generate query (Q), key (K), and value (V). The $\widehat{F}_{\text{SV}}$ features are enhanced as follows:

$$\widehat{F}_{\text{SV}} = \sigma\left(V \cdot \text{softmax}\left(Q \cdot K\right)\right) + F_{\text{SV}}. \tag{12}$$

In addition, for VHR image feature extraction, we use the popular VGG16 to extract high-level semantic features F_{VHR}.

2) Vhr-Sv Image Feature Fusion: In this study, to align and fuse the semantic features from the VHR-SV image pairs, we developed the AAFM [see Fig. 7(a)], which incorporates the Sinkhorn algorithm into cross-attention mechanism to enhance the alignment of features under different view modes so that improve the model's accuracy in identifying urban land use types [31], [47]. Specifically, this module consists of two main operations: feature alignment and feature fusion. The feature alignment operation adaptively aligns the distribution of semantic features in the feature space. First, we use a 1×1 convolution to generate a query for SV features $\widehat{F}_{\text{SV}}$, denoted as Q', and to generate a key and a value for VHR features F_{VHR}, denoted as K' and V'. Then, as shown in Fig. 7(b), the cost matrix C is calculated by the Euclidean distance between the vectors at corresponding positions in Q' and K', which can be described as

$$c_{ij} = \left\| \boldsymbol{q}_{ij} - \boldsymbol{k}_{ij} \right\|_2 \tag{13}$$

where c_{ij} represents the element at the ith row and jth column of the cost matrix C, and $\boldsymbol{q}_{ij}$ and $\boldsymbol{k}_{ij}$ are the corresponding feature vectors from Q' and K', respectively. The Sinkhorn algorithm then aligns the features by iteratively normalizing C using scaling vectors $\boldsymbol{u}^{(t+1)}$ and $\boldsymbol{v}^{(t+1)}$ to obtain the alignment-plan matrix T

$$T = \text{diag}\left(\boldsymbol{u}^{(t+1)}\right) \cdot \exp\left(-\frac{C}{\epsilon}\right) \cdot \text{diag}\left(\boldsymbol{v}^{(t+1)}\right). \tag{14}$$

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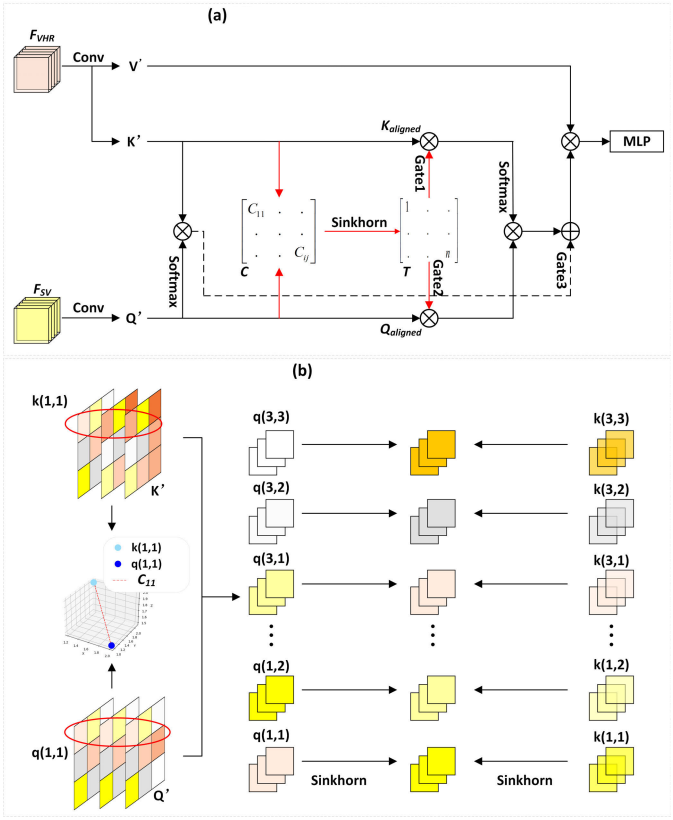


Fig. 7. (a) Details of the feature fusion module AAFM and (b) Sinkhorn algorithm adjusts the distribution of feature vectors at each pixel in the query and key.

Here, the alignment-plan matrix T represents the regularized joint probability distribution of the corresponding feature vectors in Q' and K'. Each element T_{ij} quantifies the alignment probability between the feature at position (i, j). It is then incorporated into a cross-attention mechanism to enhance the alignment of VHR-SV features F_{VHR} and $\widehat{F}_{\mathrm{SV}}$. The Q' and K' generated from F_{VHR} and $\widehat{F}_{\mathrm{SV}}$ are thus aligned as

$$Q'_{\mathrm{aligned}} = \mathrm{gate1} \cdot T \cdot Q' + \big(1 - \mathrm{gate1}\big) \cdot Q' \tag{15}$$

$$K'_{\mathrm{aligned}} = \mathrm{gate2} \cdot T \cdot K' + \big(1 - \mathrm{gate2}\big) \cdot K' \tag{16}$$

where gate1 and gate2 are learnable parameters that dynamically control each feature's contribution to alignment.

Next, the feature fusion balances the contributions of prealignment and postalignment features to land use recognition. We then use a cross-attention mechanism to fuse VHR and SV features F_{fused} defined as

$$F_{\mathrm{fused}} = V' \cdot \left(\mathrm{gate3} \cdot \mathrm{softmax}\left(\frac{Q'K'^{T}}{\sqrt{d_k}}\right) + \big(1 - \mathrm{gate3}\big) \cdot \mathrm{softmax}\left(\frac{Q'_{\mathrm{aligned}}K'^{T}_{\mathrm{aligned}}}{\sqrt{d_k}}\right)\right) \tag{17}$$

where gate3 is a learnable gating parameter. Finally, we use the fused VHR-SV features F_{fused} to determine urban land use categories.

3) Loss Function for Classification: In this study, the proposed classification model involves the calculation of two main losses associated with feature alignment and classification tasks, corresponding to align loss $\mathcal{L}_{\mathrm{align}}$ and classification loss $\mathcal{L}_{\mathrm{cls}}$. We use mean squared error (mse) to compute the align loss [48]

$$\mathcal{L}_{\mathrm{align}} = \frac{1}{N}\sum_{i=1}^{N} \left\| \boldsymbol{q}_{\mathbf{aligned},i} - \boldsymbol{k}_{\mathbf{aligned},i} \right\|_2 \tag{18}$$

where $\boldsymbol{q}_{\mathbf{aligned},i}$ and $\boldsymbol{k}_{\mathbf{aligned},i}$ represent the vectors of Q'_{aligned} and K'_{aligned} at the corresponding position i. Moreover, we use cross-entropy (CE) to compute the classification loss [49]

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{C'} y_{i,j} \log\left(\hat{y}_{i,j}\right) \tag{19}$$

where M represents the total number of samples, and C' denotes the number of classes. In total, we combine the align loss $\mathcal{L}_{\mathrm{align}}$ and classification loss L_{cls} to obtain the final loss L_{total}

$$\mathcal{L}_{\mathrm{total}} = \beta_1 \cdot \mathcal{L}_{\mathrm{cls}} + (1 - \beta_1) \cdot \mathcal{L}_{\mathrm{align}} \tag{20}$$

where β_1 is the weight between the two losses, set to 0.8 in this study based on trial and error.

D. Accuracy Assessment

For image matching, since our goal is to utilize matched image pairs for urban land use classification, we adopt matching accuracy as the sole metric to assess the final matching performance. For urban land use classification, we evaluate performance using overall accuracy (OA), precision (P), recall (R), and $F1$ score ($F1$), which together provide a comprehensive assessment of classification effectiveness

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{21}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{22}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{23}$$

$$F1 = 2 \times \frac{P \times R}{P + R} \tag{24}$$

where TP, FP, and FN indicate true positive, false positive, and false negative, respectively. In this context, positive refers to the samples with correct category predictions.

E. Implementation Details

We applied our method to two scene-based classification datasets: LU-FZ-VHR-SV and LU-ZZ-VHR-SV. In the image matching task, during training, the Adam [50] optimizer was utilized, and the learning rate was set to 0.00001 to ensure training stability and efficiency. The model was trained for 50 epochs, with the batch size set to 8. In the classification task, the model was trained for 100 epochs using the Adam optimizer with a learning rate of 0.0001 and a batch size of 16. Experiments were performed on a NVIDIA GeForce RTX 4090 GPU.

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TABLE I
CLASSIFICATION ACCURACY OF ABLATION EXPERIMENTS ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

Methods	LU-FZ-VHR-SV				LU-ZZ-VHR-SV			
	OA (%)	P (%)	R (%)	F1 (%)	OA (%)	P (%)	R (%)	F1 (%)
a (without SV images)	87.50	87.74	87.50	87.18	95.00	93.90	95.83	94.88
b (with SV images)	**91.00**	**90.80**	**91.50**	**91.11**	**98.00**	**97.62**	**98.33**	**97.93**

F. Performance Evaluation of Image Matching and Urban Land Use Classification

1) Evaluation of the Effectiveness of Sv Images in Urban Land Use Classification: To evaluate the effectiveness of SV images in urban land use classification, we conducted ablation experiments with and without incorporating SV images. Specifically, we first applied our proposed classification method, which integrates features from both VHR and SV images. To evaluate the specific contribution of SV images, we then removed the SV branch and performed classification using only VHR images.

2) Comparison of Different Correlation Calculation Methods: To assess the effectiveness of the proposed SCCC, we conducted a comparative analysis with other correlation calculation methods, including dynamic similarity matching (DSM) [51] and Zhang's method [52]. More specifically, DSM calculates the correlation between VHR and SV image features by aligning them along the horizontal direction using a sliding window and point product. The Zhang's method calculates the correlation between images based on the angle between their feature vectors.

3) Evaluation of the Effectiveness of Feature Alignment in Feature Fusion: To evaluate the effectiveness of feature alignment in feature fusion, we compared the AAFM with three fusion methods, corresponding to feature concatenation, cross-attention, and cross-attention with position encoding [53]. While both methods are widely applied in feature fusion tasks, we focused on their limitations in handling feature misalignment between images from different view modes. Specifically, we investigated how these traditional methods perform without alignment considerations, and compared the impact of spatially aligned features using position encoding on feature fusion. By comparing these established approaches with AAFM, we aim to highlight the distinct advantages of our method in addressing feature alignment and improving feature fusion across different view modes.

4) Comparison With Existing Methods for Urban Land Use Classification: So far, only a limited number of deep learning methods have been specifically designed to combine SV and VHR Images for urban land use classification. In this study, to evaluate the effectiveness of the proposed classification method, we compared it with three state-of-the-art (SOTA) methods in other VHR image classification domains, corresponding to a CNN-based method, AGOS [54], a transformer-based method, CCT [55], and a CNN-transformer-based method, DecoupleNet [56]. AGOS utilizes a CNN to extract multigrain feature representations from VHR images, integrating them through a semantic fusion module. CCT combines convolutional tokenization with transformers, allowing for efficient feature extraction while maintaining a compact model size. DecoupleNet adopts the feature decoupling, which divides features into multiple branches for processing, allowing each branch to be optimized for different tasks or scales.

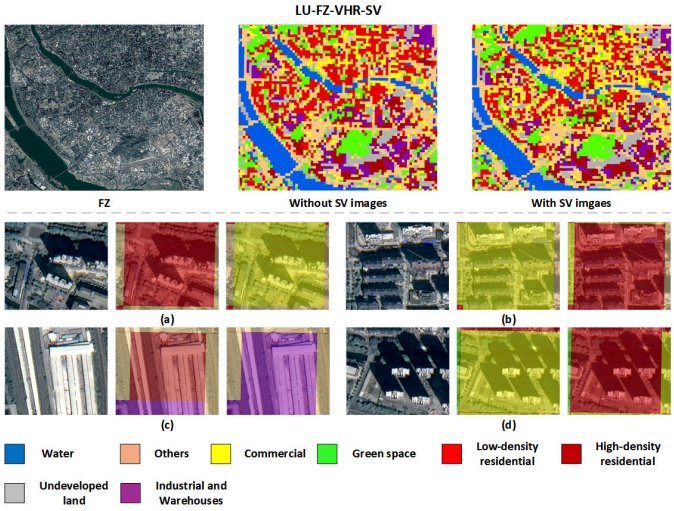


Fig. 8. Urban land use classification results with and without using SV images on the LU-FZ-VHR-SV dataset. (a)–(d) Illustrate the detailed results. From left to right in each row are the ground truth, the result without SV images, and the result with SV images.

IV. EXPERIMENTS

A. Evaluation of the Effectiveness of Sv Images in Urban Land Use Classification

We conducted ablation studies to assess the effectiveness of SV images. Table I lists the classification accuracy with and without using SV images on the LU-ZZ-VHR-SV and LU-FZ-VHR-SV datasets. Figs. 8 and 9 show the urban land use classification results with and without using SV images. Our results indicate that the use of SV images improves the accuracy of urban land use classification, particularly for categories with high interclass similarity, such as low-density residential and commercial areas, as well as industrial and high-density residential areas. Specifically, on the LU-FZ-VHR-SV dataset, it outperforms only using VHR images by 3.5%. On the LU-ZZ-VHR-SV dataset, the classification accuracy improves by 3%.

B. Comparison of Different Correlation Calculation Method for Images Matching

We compared the proposed SCCC with two correlation calculation methods, including DSM and Zhang's method.

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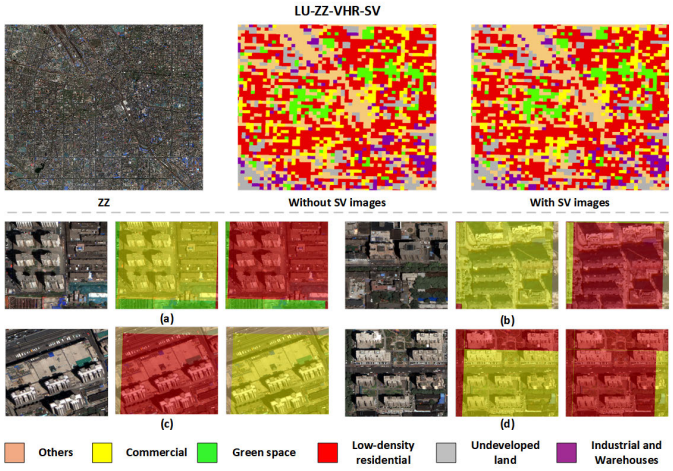


Fig. 9. Urban land use classification results with and without using SV images on the LU-ZZ-VHR-SV dataset. (a)–(d) Illustrate the detailed results. From left to right in each are the ground truth, the result without SV images, and the result with SV images.

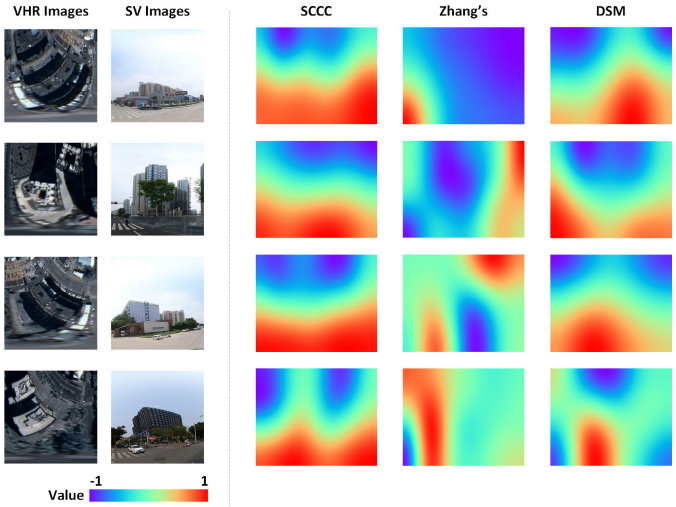


Fig. 10. Examples of correlation scores for different methods applied to the matched images. These present the pixel-level feature values extracted from the image pairs, with the visualization reflecting.

TABLE II
MATCHING ACCURACY ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

Datasets	Methods	Matching accuracy (%)
LU-FZ-VHR-SV	Zhang's	73.76
	DSM	76.25
	SCCC	**80.50**
LU-ZZ-VHR-SV	Zhang's	70.42
	DSM	72.22
	SCCC	**73.93**

Table II lists the matching accuracy of these experiments using different correlation calculation methods on the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets. Results show that SCCC improves the matching accuracy. Specifically, on the LU-FZ-VHR-SV dataset, the matching accuracy reached 76.25% using the DSM, outperforming Zhang's method. Furthermore, integrating the point product with SCCC improves the matching accuracy to 80.5%. On the LU-ZZ-VHR-SV datasets, using SCCC also demonstrates a relative advantage, with a matching accuracy of 73.93%, outperforming the DSM and Zhang's method, further validating the robustness and effectiveness of SCCC across different datasets.

In addition, we visualized the correlation scores learned for each correlation calculation methods. Fig. 10 shows the correlation scores for different correlation calculation methods on the LU-FZ-VHR-SV dataset. The red and blue regions represent areas with high- and low-correlations, respectively. These figures demonstrate that the SCCC effectively captures the correlation between VHR and SV images, with the component focusing more on key features in high-correlation areas and less on secondary features in low-correlation areas. Fig. 11 shows a schematic of the matched VHR and SV images, which indicates that the proposed matching method can accurately identify correct correspondences across diverse urban scenes.

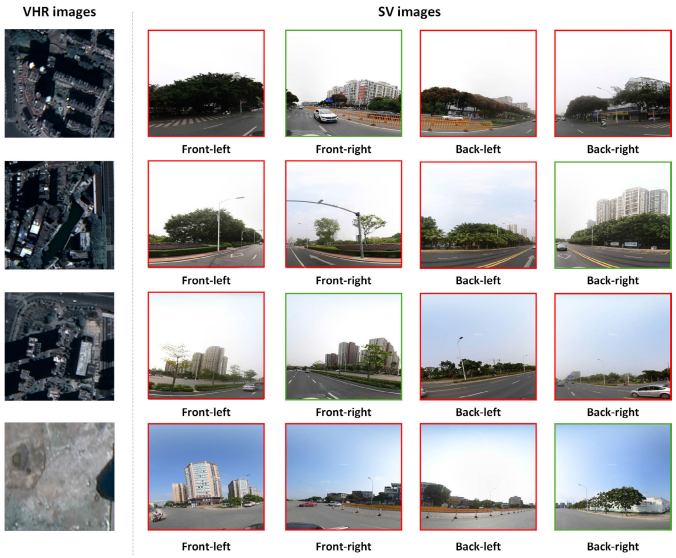


Fig. 11. Examples of matching results, with red boxes denoting mismatches and green boxes representing correctly matched regions.

C. Effectiveness of Feature Alignment in Feature Fusion

To explore the impact of the feature alignment in fusing images from different view modes, we compared the proposed AAFM with three traditional methods of feature fusion. Qualitative and quantitative assessments were performed to ensure a comprehensive evaluation of the module. Table III lists the classification accuracy of different fusion methods on the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets. The results show that the AAFM improves the classification performance. Specifically, on the LU-FZ-VHR-SV dataset, the OA with AAFM outperforms feature concatenation, cross-attention, and cross-attention with position encoding by 2.5%, 21%, and 21%. On the LU-ZZ-VHR-SV dataset, it improves by 1%, 3%, and 27%, respectively.

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TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT FEATURE FUSION METHODS ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

Datasets	Methods	OA (%)	P (%)	R (%)	F1 (%)
LU-FZ VHR-SV	Concatenation	88.50	88.85	88.85	88.62
	Cross-attention	70.00	69.32	70.00	68.62
	Cross-attention (Position encoding)	70.00	79.79	70.00	69.21
	AAFM	**91.00**	**90.80**	**91.50**	**91.11**
LU-ZZ VHR-SV	Concatenation	97.00	96.51	97.50	96.91
	Cross-attention	95.00	95.47	94.17	94.72
	Cross-attention (Position encoding)	71.00	70.10	67.92	68.34
	AAFM	**98.00**	**97.62**	**98.33**	**97.93**

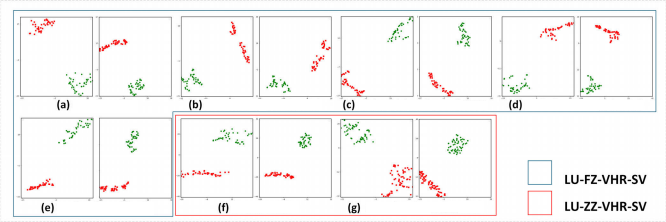


Fig. 12. Overview of the T-SNE visualization of queries and keys in different land use categories from two datasets, with 40 samples selected from each category. On the left side of each image is the result generated without Sinkhorn's cross-attention, and on the right side is the result using AAFM. The query (green) and key (red) from the same category tend to be closer in feature space. (a) and (f) Commercial. (b) and (g) Low-density residential. (c) Industrial and warehouse. (d) High-density residential. (e) Undeveloped land.

Fig. 12 shows the distribution of the query and key in the feature space generated by cross-attention, comparing the cases with and without Sinkhorn alignment. This figure visually demonstrates that Sinkhorn alignment enhances the clustering of query and key in the feature, while Table III shows that this enhanced alignment leads to higher classification accuracy. This enhanced alignment, in turn, contributes to improved classification performance.

D. Comparison of Different Urban Land Use Classification Methods

We compared our classification method with AGOS, CCT, and DecoupleNet on the LU-ZZ-VHR-SV and LU-FZ-VHR-SV datasets. Table IV shows that our classification method achieved the highest OA of 91% on the LU-ZZ-VHR-SV dataset and 98% on the LU-FZ-VHR-SV dataset. It improved OA by 7%, 6.5%, and 1.5% over AGOS, CCT, and DecoupleNet on the LU-ZZ-VHR-SV dataset, and by 4%, 15%, and 3% on the LU-FZ-VHR-SV dataset. These results underscore the effectiveness of our proposed classification method for urban land use classification. Figs. 13 and 14 show the urban land use classification results using the four methods on both datasets. The classification results by our method have shown that SV images significantly improve urban land use classification, particularly in distinguishing low-density areas and commercial areas.

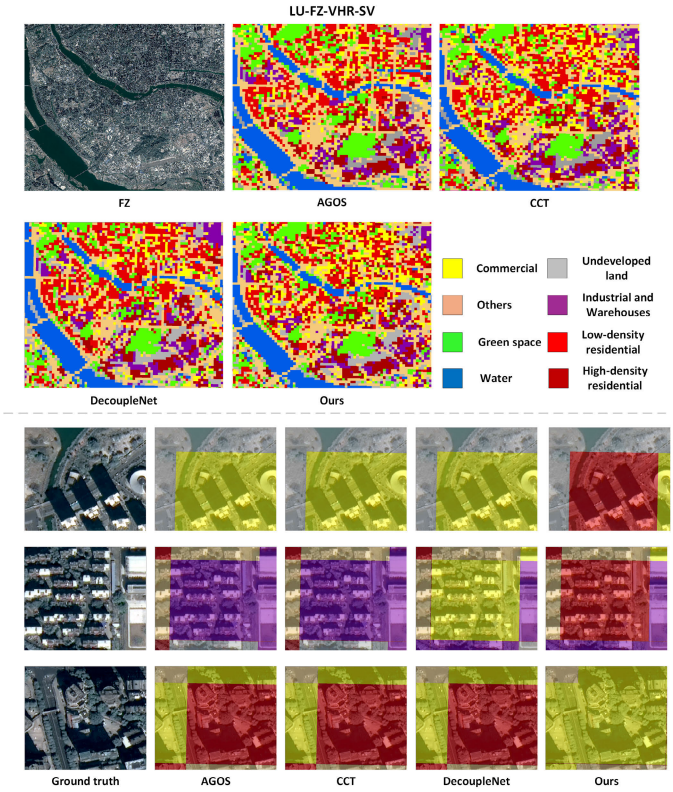


Fig. 13. Urban land use classification results using different methods on the LU-FZ-VHR-SV dataset. (Top to bottom) The ground truth corresponds to low-density residential, low-density residential, and commercial areas. Our method successfully identifies all three categories.

TABLE IV

URBAN LAND USE CLASSIFICATION RESULTS BY DIFFERENT CLASSIFICATION METHODS ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

Datasets	Methods	OA (%)	P (%)	R (%)	F1 (%)
LU-FZ VHR-SV	AGOS	84.00	83.70	84.50	84.17
	CCT	84.50	84.50	84.50	84.51
	DecoupleNet	89.50	89.22	89.50	89.36
	Ours	**91.00**	**90.08**	**91.50**	**91.11**
LU-ZZ VHR-SV	AGOS	94.00	92.78	95.00	93.88
	CCT	83.00	82.33	83.00	82.77
	DecoupleNet	95.00	93.95	95.83	94.88
	Ours	**98.00**	**97.62**	**98.33**	**97.93**

E. Effectiveness of Combined Loss Function

To evaluate the effectiveness of combined loss functions, we conducted comparative experiments with three loss functions on LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets. Table V

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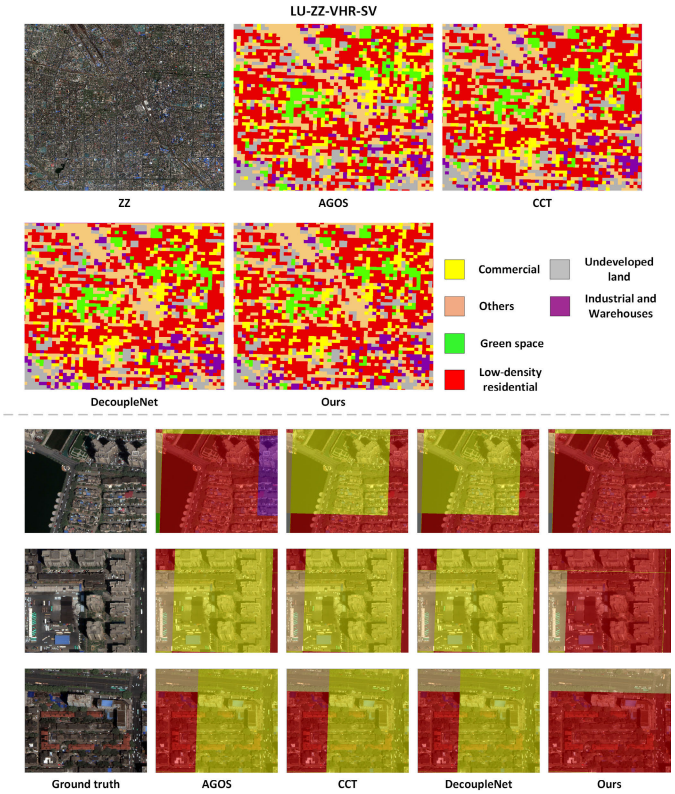


Fig. 14. Urban land use classification results using different methods on the LU-ZZ-VHR-SV dataset. (Top to bottom) The ground truth corresponds to low-density residential, low-density residential, and low-density residential. Our method successfully identifies all three categories.

TABLE V

CLASSIFICATION ACCURACY OF DIFFERENT LOSS FUNCTIONS ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

Datasets	Methods	OA (%)	P (%)	R (%)	F1 (%)
LU-FZ VHR-SV	CE_{CS}	87.50	89.95	87.50	86.56
	CE	**91.00**	**91.12**	91.00	91.02
	CE_{MSE}	**91.00**	90.80	**91.50**	**91.11**
LU-ZZ VHR-SV	CE_{CS}	95.00	94.98	94.58	94.77
	CE	96.00	95.83	95.83	95.83
	CE_{MSE}	**98.00**	**97.62**	**98.33**	**97.93**

lists the classification accuracy. Our results show that the proposed loss function is robustness and reliability for urban land use classification. Specifically, the use of CE loss alone achieved an OA of 91.00%, with P, *R*, and *F1* of 91.12%, 91.00%, and 91.02%, respectively. When mse was used to align, the OA remained at 91.00%, but the *F1* improved slightly to 91.11%. Notably, mse contributed to a significant improvement in *R*, increasing it to 91.50%, demonstrating its effectiveness in optimizing feature fusion through feature alignment. In contrast, the cosine similarity (CS) loss combined with CE exhibited weaker performance for feature

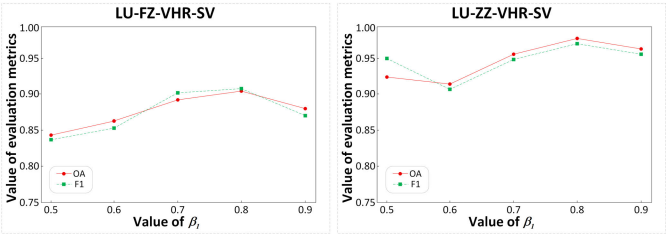


Fig. 15. Effect of the loss weight β_1 value on urban land use classification metrics on the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets.

fusion, achieving an OA of only 87.50% and an *F1* of 86.56%. This disparity is likely due to CS loss prioritizing directional consistency while neglecting the magnitude information of features.

In addition, to assess the influence of the different loss weight β_1 values for the urban land use classification, we conducted a sensitivity analysis. The results, shown in Fig. 15, indicating when β_1 is assigned a value of 0.8 yields the best performance on both the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets.

V. DISCUSSION

In this study, we investigated the use of SV images to assist VHR images for urban land use classification at the scene level. The results demonstrate that the proposed matching and classification method is effective. The use of SV images enhances the recognition of complex urban functional areas, particularly for urban land use units with high inter-class consistency. Furthermore, our proposed SCCC improves matching performance, minimizing the mismatches caused by the complexities of urban environments. Most existing feature fusion mechanisms rely on feature concatenation or cross-attention, with only a few studies exploring the distribution within the feature space. In this study, we developed the AAFM, which optimizes the distribution of different view-mode features in the feature space by integrating the Sinkhorn algorithm, thereby enhancing the urban land use classification accuracy.

A. Contributions of Proposed Image Matching Method

There are two main challenges in matching SV images with VHR images, i.e., most SV images lack a reference direction, and there is a significant view modes disparity between SV and VHR images. These issues complicate the matching process. To address these challenges, we first simulate the capture direction of the SV images using the spatial position constraints, establishing a coarse-level correspondence. We then introduce a viewpoint transformation to adjust the view mode of the VHR images, eliminating the view-mode differences between the SV and VHR images. This proposed matching process reduces the matching scope while enabling the deep neural network to learn correlations from image pairs.

B. Contributions of Strip-Layered Strategy to the Image Matching

While using DSM [51] and Zhang's method [52] is efficient and convenient. However, these correlation calculation

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TABLE VI
COMPARISON OF CLASSIFICATION ACCURACY USING DIFFERENT SV FEATURE EXTRACTION METHODS ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

SV feature extraction	LU-FZ-VHR-SV				LU-ZZ-VHR-SV			
	OA (%)	P (%)	R (%)	F1 (%)	OA (%)	P (%)	R (%)	F1 (%)
ResNet50	84.00	85.54	84.00	83.94	96.00	95.83	95.83	95.83
VGG16	86.00	86.39	86.00	85.93	95.00	95.47	94.17	94.72
Swin Transformer	83.35	83.68	83.50	83.39	93.00	92.42	93.75	92.84
FastSAM	**91.00**	**90.80**	**91.50**	**91.11**	**98.00**	**97.62**	**98.33**	**97.93**

TABLE VII
COMPARISON OF CLASSIFICATION ACCURACY USING DIFFERENT VHR FEATURE EXTRACTION METHODS ON THE LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

VHR feature extraction	LU-FZ-VHR-SV				LU-ZZ-VHR-SV			
	OA (%)	P (%)	R (%)	F1 (%)	OA (%)	P (%)	R (%)	F1 (%)
Swin Transformer	90.00	90.11	90.00	89.97	97.00	96.71	97.08	96.89
FastSAM	88.00	89.90	89.50	89.79	85.00	85.71	90.00	87.80
ResNet50	88.50	87.89	88.00	87.80	96.00	95.45	96.67	95.89
VGG16	**91.00**	**90.80**	**91.50**	**91.11**	**98.00**	**97.62**	**98.33**	**97.93**

methods are often applied to relatively simple and open environments, and inevitably have limitations in complex urban settings. Therefore, we proposed the SCCC, which employs a strip-layered strategy to calculate the correlation between corresponding image strips and dynamically adjusts them using an attention mechanism. The strip-layered strategy fully considers the detailed information between images. Experimental results show that SCCC clearly focuses on local details during the matching process (see Fig. 10). Compared to directly calculating correlations, it achieves better matching accuracy of image pairs in complex backgrounds.

C. Contributions of Feature Alignment in Feature Space to Feature Fusion

In this study, we found that integrating the Sinkhorn algorithm into the cross-attention module for better feature alignment during fusion improves urban land use classification accuracy. Recent studies have attempted to enhance the effectiveness of the self-attention mechanism by aligning features in the feature space [57]. We apply this approach to cross-attention, aligning feature distributions in the feature space to enhance the effectiveness of feature fusion. As shown in Fig. 12, AAFM visualizations confirm the effectiveness of feature alignment. Furthermore, to explore the impact of spatial alignment on feature fusion, we introduced position encoding into the cross-attention mechanism. The results (see Table III), particularly on the LU-ZZ-VHR-SV dataset, indicate that position encoding can degrade fusion performance. This demonstrates that absolute spatial position cues become less relevant when there are significant differences between view modes. In this condition, aligning features in the feature space is the necessary operation for feature fusion.

D. Sv Features Extracted by Our Modified FastSAM

In this study, we modified FastSAM to extract features from SV images. Trained on large-scale real-world data, the model demonstrates superior capability in capturing fine-grained, context-aware representations from complex street-level scenes. As shown in the ablation study (see Table VI), FastSAM consistently outperforms other mainstream SV feature extractors [58], [59]. Its segmentation-driven architecture enables robustness to occlusion, visual clutter, and viewpoint variation, which is particularly valuable for urban land use classification, where semantic richness and structural complexity are prevalent. These results indicate that SAM and its variants are strong alternatives to traditional backbones in SV-based tasks. Future work may explore their integration into multimodal or semi-supervised frameworks, especially under sparse SV coverage or limited labeled data.

E. Vhr Features Extracted by VGG16

In this study, we chose VGG16 for feature extraction from VHR images. To explore the rationale behind combining this architecture with FastSAM, we conducted ablation experiments. As shown in Table VII, the combination of VGG16 for VHR image feature extraction and FastSAM for SV image feature extraction yielded the best classification performance. We believe that simple feature extraction methods may be more effective due to the nature of the datasets. Relatively simple VHR datasets may not require complex feature extraction architectures.

F. Applicability of Sv Images in Urban Land Use Classification

We have demonstrated the contribution of SV images to urban land use classification by comparing results with and without their inclusion on the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets (see Table I). However, the overall impact of our method which first matching image pairs and then using the matched pairs for classification has not yet been fully investigated. To evaluate this impact, we generate two land

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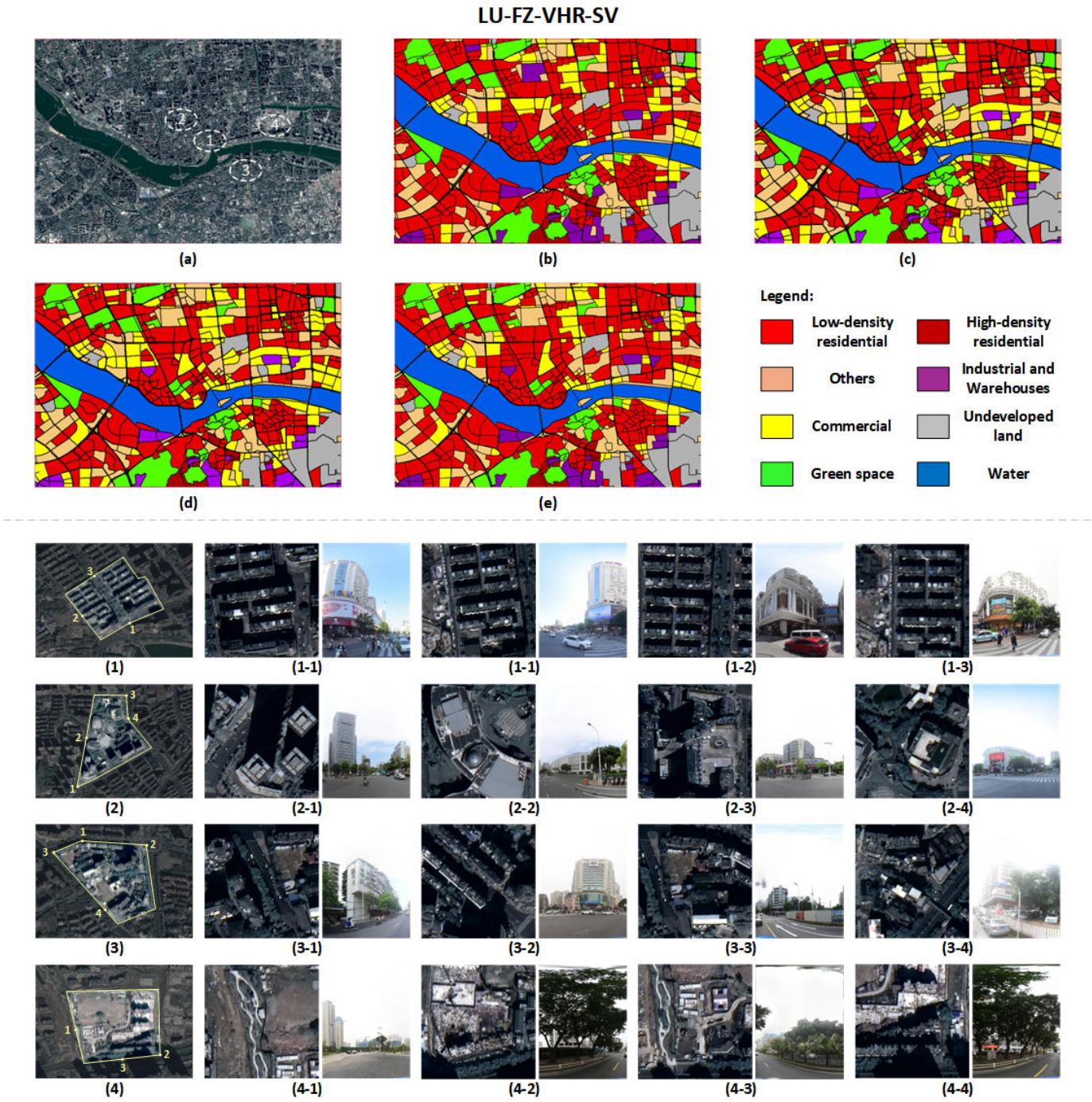


Fig. 16. (a) Shows the original VHR images of Fuzhou; (b) classification results without SV images; (c) with full mapping; (d) with partial mapping; and (e) with minimal mapping. The last also provides a detailed comparison of selected regions, highlighting differences between (b) and (d). (1)–(4) present detailed urban land use classification results, offering partial views that support subsequent analysis.

use maps on Fuzhou and Zhengzhou (see Figs. 16 and 17). Specifically, to generate a land use map from a large VHR image, we apply a sliding window cropping to extract regular patches and classify them using VGG16. In parallel, the VHR-SV image pairs, constructed based on the road network, are classified using the proposed method. The classification results are then projected onto the nearest sliding-window-cropped VHR patches, which further contributes to the regular block-like appearance in the final prediction maps. To regulate this mapping, we define three constraints: 1) full mapping, using all SV-based predictions; 2) partial mapping, excluding undeveloped and high-density residential areas; and 3) minimal mapping, removing SV images with excessive tree or sky coverage.

Tables VIII and IX present the classification results after full mapping of the SV-matched VHR images and solely on VHR images. Meanwhile, we also compiled the number of correct minimal mapping, as shown in Table X, which demonstrates

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TABLE VIII

DEMONSTRATION OF SPECIFIC CASES WHERE THE RESULTS OF FULL MAPPING ON THE LU-FZ-VHR-SV DATASET. THE RED TEXT INDICATES INCORRECT RESULTS

Datasets	**LU-FZ-VHR-SV**					
Category	Commercial	Commercial	Low-density residential	Low-density residential	Industrial and Warehouses	Low-density residential
VHR images						
SV images						
VHR images Results	Low-density residential	Low-density residential	Industrial and Warehouses	Industrial and Warehouses	Industrial and Warehouses	Commercial
SV-matched VHR Results	Commercial	Commercial	High-density residential	Low-density residential	Commercial	Low-density residential

TABLE IX

DEMONSTRATION OF SPECIFIC CASES WHERE THE RESULTS OF FULL MAPPING ON THE LU-ZZ-VHR-SV DATASET. THE RED TEXT INDICATES INCORRECT RESULTS

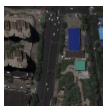
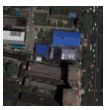
Datasets	**LU-FZ-VHR-SV**					
Category	Commercial	Commercial	Low-density residential	Low-density residential	Industrial and Warehouses	Low-density residential
VHR images						
SV images						
VHR images Results	Low-density residential	Low-density residential	Industrial and Warehouses	Industrial and Warehouses	Industrial and Warehouses	Commercial
SV-matched VHR Results	Commercial	Commercial	High-density residential	Low-density residential	Commercial	Low-density residential

TABLE X

STATISTICS OF MINIMAL MAPPING IN LU-FZ-VHR-SV AND LU-ZZ-VHR-SV DATASETS

Datasets	**Number of sliding-window patches**	**Number of minimal mapping**	**Number of correct minimal mapping**
LU-FZ-VHR-SV	6720	190	127
LU-ZZ-VHR-SV	3204	111	84

the effectiveness of our method in urban land use classification across the entire city.

Specifically, in practical applications, the use of SV images still requires some tradeoffs. As shown in Tables VIII and IX: SV images are challenging for distinguishing high-density from low-density residential areas due to the structural similarity between high-density areas and older low-density ones, making identification difficult. Undeveloped areas are often surrounded by walls, hindering access to SV images. Despite these challenges, SV images have proven effective for identifying commercial areas and low-density residential areas. Figs. 16 and 17 provide more detailed presentations. For example, Figs. 16(d) and (e) and 17(d) and (e) show that as the number of SV images with heavy occlusions decreases, the amount of commercial land in the prediction results also decreases. This is because the reduction in image matching errors leads to a final urban land use classification that more closely aligns with real-world conditions. Fig. 16 areas (1) and (2) depict two commercial areas, but in the VHR images, they are similar to medium to low-density residential areas. By incorporating SV images, these areas were accurately identified. Fig. 16 area (4) shows an area surrounded by trees; after limiting occlusion, the correct classification was achieved. In contrast, Fig. 17 area (1) illustrates that as long as there is no complete occlusion, SV images can assist in obtaining the correct classification. In conclusion, as shown in Table X, when factors like category types, occlusions were controlled, SV-assisted classification significantly improved the identification of commercial and low-density residential areas. Nonetheless, future studies can quantitatively assess the contribution of SV images to urban land use classification by

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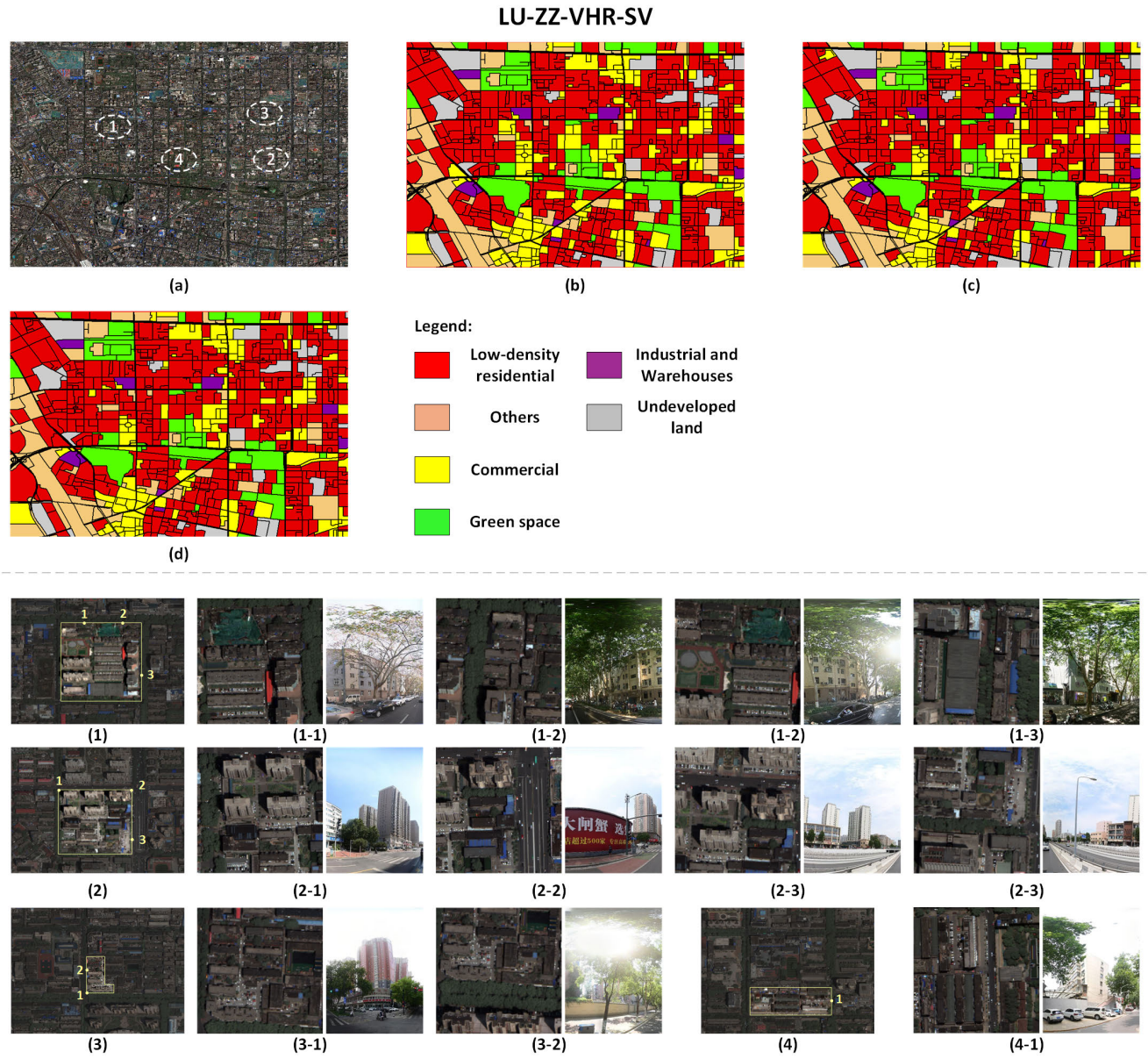


Fig. 17. (a) Shows the original VHR images of Zhengzhou; (b) classification results without SV images; (c) with full mapping; and (d) with minimal mapping. (1)–(4) present detailed urban land use classification results, offering partial views to support subsequent analysis.

analyzing the proportion of different objects present in the SV scenes.

Moreover, the errors in the first-stage matching may affect the final classification accuracy. The proposed matching process is performed in two levels: a coarse-level matching based on spatial position constraints and viewpoint transformation, followed by a fine-level matching using a deep neural network and SCCC. At the coarse-level, the matching error is minimal. The main matching errors occur at the fine-level, primarily due to two factors:

1) The image quality of the SV images. For example, significant occlusions, such as trees, or a large proportion of sky in SV images when vehicles are driving through open areas, can hinder the performance of the deep neural network. In these cases, the network may fail to effectively learn representative features, leading to an inability to correctly calculate the correlation between the SV images and the VHR images. To be honest, the SV images collected by vehicles inevitably contain some ill-posed images, which are related to image quality issues. A straightforward approach to address this data quality problem is to preprocess the data and filter out low-quality SV images. When we filter out low-quality SV images (with tree or sky coverage exceeding 50%), we observed a noticeable reduction in the over representation of commercial areas in the urban land use maps (see Figs. 16 and 17), resulting in outputs that better reflect realistic conditions.
2) The model accuracy. In this study, despite our proposed method achieved promising results (e.g., Fig. 10, Table II, comparing the proposed method with Zhang's

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and DSM), we acknowledge that more advanced deep learning method to enhance feature learning may contribute to the improvement of fine-level matching, e.g., the future work can involve using more advanced feature extractors and novel correlation computation methods to improve matching accuracy. We believe that such advancement will further improve the land use classification accuracy based on image matching.

VI. Conclusion

In this study, we proposed a novel method for matching VHR images with SV images for urban land use classification. In image matching, we established a coarse spatial alignment between VHR and SV images based on spatial position constraints, and bridged view-mode differences by leveraging the viewpoint transformation. To mitigate the impact of complex urban environments on the image matching, we designed an SCCC to model the correlation between image pairs in a strip-wise manner. For the matched image pairs, we developed an AAFM to align features from different view modes in the feature space, thereby facilitating more effective feature fusion. We conduct urban land use classification experiments on two customized datasets, LU-FZ-VHR-SV and LU-ZZ-VHR-SV. Based on the results, we draw the following conclusions.

1) The use of SV images outperforms only use VHR images for urban land use classification on both LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets, with OA improvements of 3.5% and 3.0%, respectively.
2) The strip-based correlation calculation method, combined with attention-based dynamic adjustment, effectively mitigates mismatches arising from the complexity of urban environments.
3) Aligning features in the feature space significantly improves the effectiveness of feature fusion, particularly when dealing with images captured from different view modes.
4) The proposed classification method outperforms existing models (e.g., AGOS, CCT, and DecoupleNet) in urban land use classification on the LU-FZ-VHR-SV and LU-ZZ-VHR-SV datasets, achieving OA of up to 91% and 98%, respectively.
5) While the overall contribution of SV images to urban land use classification is limited, it proves particularly beneficial in distinguishing commercial land, high-density residential areas, and low-density residential areas.

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Zheyao Yu received the B.Sc. degree from Zhejiang A&F University, Hangzhou, China, in 2022. He is currently pursuing the M.Sc. degree in surveying and mapping engineering with the Academy of Digital China, Fuzhou University, Fuzhou, China.

His research interests lie in machine learning and intelligence in remote sensing, remote sensing classification, and cross-view localization.



Mengmeng Li (Member, IEEE) received the M.Sc. degree in cartography and geography information systems from Fuzhou University, Fuzhou, China, in 2012, and the Ph.D. degree from the Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, Enschede, The Netherlands, in April 2017.

He was a Post-Doctoral Researcher with ITC from April 2017 to October 2019 and an Exchange Ph.D. Researcher with The University of Oklahoma, Norman, OK, USA, in 2016. He is currently an Associate Professor with the Academy of Digital China (Fujian), Fuzhou University. His research interests include machine learning and intelligence in remote sensing, mapping, and change detection of land cover and land use, time-series image analysis, and uncertainty handling for remote sensing image analysis.

Dr. Li's Ph.D. Thesis titled "Urban Land Use Extraction From Very High Resolution Remote Sensing Images" was awarded the EuroSDR Best Ph.D. Thesis Award in 2018.



Xiangtao Zheng (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees in signal and information processing from Chinese Academy of Sciences, Xi'an, China, in 2014 and 2017, respectively.

He is currently a Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China. His main research interests include computer vision and pattern recognition.

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